

RICAN: A Unified Framework for Optimal Fixed-Rate Source Coding via Diophantine Resonance

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Abstract

This paper presents a unified mathematical framework for constructing asymptotically optimal fixed-length binary representations for sequences over finite alphabets. The core contribution is a systematic method, based on Diophantine approximation via continued fractions, for selecting block sizes that achieve vanishingly small representation overhead for any alphabet size B .

For an alphabet of size B , the minimum fixed-length representation for blocks of length K requires $n(K) = \lceil K \log_2(B) \rceil$ bits. The redundancy $R(K) = n(K) - K \log_2(B)$ is substantially reduced along block sizes derived from continued fraction convergents of $\log_2(B)$. The convergents of $\log_2(B)$ provide block sizes for which the asymptotic efficiency $\eta(K) = K \log_2(B) / \lceil K \log_2(B) \rceil$ approaches 1.

For the decimal case ($B = 10$), we identify a family of block sizes that achieve near-perfect efficiency. The block $K = 783,032$ decimal digits is represented in exactly 325,148 bytes (2,601,184 bits), with a per-symbol redundancy of only 6.64×10^{-13} bits. This represents an improvement of over five orders of magnitude compared to the previously known convergent 2136/643, and places the coding efficiency within $10^{-9}\%$ of the Shannon limit, demonstrating that fixed-length coding can achieve near-perfect efficiency without statistical modeling.

Unlike statistical compression methods, this approach does not exploit source redundancy. It provides a deterministic, reversible mapping whose output size depends only on the input length and alphabet size, not on statistical properties. We identify a complete hierarchy of optimal block sizes arising from the continued fraction expansion of $\alpha = \log_2(10)/8$, establishing the theoretical connection between number-theoretic resonance and the Shannon limit. Validation on up to 10^9 decimal digits confirms the theoretical efficiency and linear computational complexity for fixed block size.

Index Terms

Fixed-rate source coding, radix conversion, finite alphabets, continued fractions, Diophantine approximation, Shannon limit, optimal block sizes, block encoding.

I. INTRODUCTION

Lossless source coding seeks reversible representations whose average description length approaches the entropy of the source while preserving complete recoverability. Shannon's Noiseless Coding Theorem establishes that no uniquely decodable code can achieve an expected rate below the source entropy [1]. Since its publication, numerous coding techniques, including Huffman coding, arithmetic coding, and dictionary-based methods, have been developed to approach this theoretical limit under different assumptions regarding source statistics.

When the source alphabet consists of B equally likely symbols, the per-symbol entropy is

$$H = \log_2(B)$$

bits per symbol. Unlike binary alphabets ($B = 2$), this quantity is generally irrational for non-power-of-two alphabets. Consequently, no fixed-length binary representation can achieve exactly the entropy rate on a per-symbol basis. Any practical fixed-rate encoder must therefore operate on blocks of symbols, introducing a small representation overhead determined by the block length.

The challenge of achieving the Shannon limit with fixed-length codes has remained open since Shannon's original work. While variable-length codes such as arithmetic coding can asymptotically achieve the entropy rate, fixed-length codes have traditionally been limited by rounding inefficiencies.

This observation motivates the present work. Rather than constructing a statistical compressor, we investigate the general representation problem: given every possible block of K symbols from an alphabet of size B , what is the smallest fixed binary representation capable of encoding all such blocks bijectively? The answer is

$$n(K) = \lceil K \log_2(B) \rceil$$

bits, a direct consequence of the pigeonhole principle.

The method is named RICAN (Representational Isomorphic Coding for Arbitrary Numeration), reflecting its objective of constructing deterministic bijective representations between finite-alphabet blocks and fixed-length binary words.

The principal contribution of this paper is the observation that continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. Specifically, we show that the convergents of the continued fraction expansion of $\log_2(B)$ provide block sizes that achieve highly efficient rounding. This connection between Diophantine approximation and fixed-length source representation is the main result.

For the decimal case $B = 10$, the convergent

$$\log_2(10) \approx \frac{2136}{643}$$

permits every block of 643 decimal digits to be represented in exactly 267 bytes, with a per-symbol ceiling redundancy of

$$\frac{\varepsilon}{643} \approx 3.65 \times 10^{-7}$$

bits, where $\varepsilon = 2136 - 643 \log_2(10) \approx 2.35 \times 10^{-4}$ is the block overhead. Among the first convergents, 643 is a denominator yielding an approximation error below 10^{-6} .

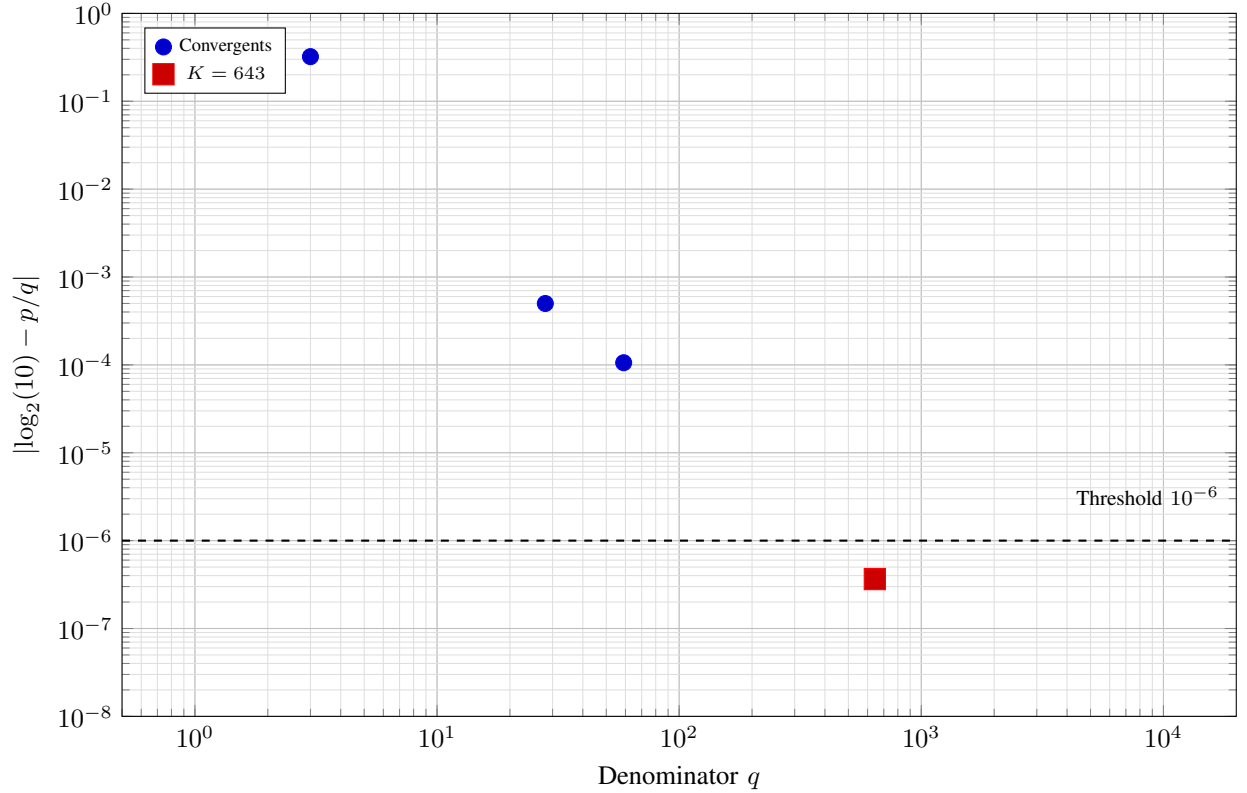


Fig. 1. Approximation error of the convergents of $\log_2(10)$ in logarithmic scale. The convergent 2136/643 produces a dramatic reduction in error, dropping from approximately 10^{-4} to 10^{-7} . The dashed line indicates the 10^{-6} threshold; 643 is the first denominator that falls below this threshold. The higher-order convergent leading to $N = 783,032$ achieves an error of approximately 10^{-13} , placing it far below the scale shown.

This paper makes three principal contributions:

- 1) A general methodology for selecting highly efficient block sizes for fixed-length representation of any finite alphabet using continued fraction convergents of $\log_2(B)$.
- 2) A proof that continued-fraction convergents provide a systematic sequence of block sizes with remarkably small representation overhead and asymptotic efficiency approaching one.
- 3) A practical instantiation for decimal digits ($B = 10$) with $K = 643$ and its experimental validation, along with the discovery of a hierarchy of optimal block sizes via Diophantine resonance.

The remainder of the paper is organized as follows. Section II reviews related work. Section III presents the general mathematical formulation. Section IV establishes the efficiency of convergent-based block selection. Section V presents validation results for the decimal case.

II. RELATED WORK

The problem of representing information with the minimum possible number of bits has been a central topic in information theory since Shannon's seminal work [1]. Existing approaches can be broadly classified into statistical source coding methods, dictionary-based methods, and fixed-length representation techniques.

A. Statistical Source Coding

Statistical source coding assigns shorter codewords to more probable symbols to minimize expected code length. Huffman coding [5] achieves optimal prefix codes for known symbol probabilities. Arithmetic coding [7], [11] removes integer-length restrictions by representing sequences as subintervals of the unit interval. Asymmetric numeral systems (ANS) [13] provide comparable efficiency with lower computational cost. These approaches optimize expected description length based on source statistics.

B. Dictionary-Based Methods

Dictionary methods, beginning with Lempel–Ziv algorithms [6], exploit repeated substrings through dictionary references. Modern coders such as gzip, bzip2, and xz combine this with entropy coding [9], [10], [12]. These methods are effective when redundancy exists in the source.

C. Fixed-Length Representations

Fixed-length representation of finite alphabets is a classical problem. Given an alphabet of cardinality B , every symbol requires $\lceil \log_2(B) \rceil$ bits. For blocks of length K , the minimum fixed-length representation requires $\lceil K \log_2(B) \rceil$ bits. Enumerative source encoding [14] provides systematic ranking and unranking methods. Mixed-radix representations [15] generalize this to varying radices.

D. Position of RICAN

RICAN differs from statistical methods by not exploiting source statistics. It provides a deterministic fixed-length representation whose output size depends only on input length and alphabet size. The contribution is not the existence of the representation, but the systematic selection of block sizes via continued fraction theory.

III. REPRESENTATION VERSUS STATISTICAL CODING

It is important to distinguish representation optimization from statistical coding.

A statistical coder reduces average description length by exploiting redundancy. Its performance depends on source statistics.

In contrast, RICAN does not modify information content or reduce entropy. It provides the smallest fixed-length binary representation for a given block size, independent of source statistics.

IV. GENERAL MATHEMATICAL FORMULATION

A. Alphabet and Block Space

Let $\Sigma_B = \{0, 1, \dots, B-1\}$ be an alphabet of size $B \geq 2$. For every positive integer K ,

$$\Sigma_B^K$$

denotes the set of all strings of length K over this alphabet, with cardinality

$$|\Sigma_B^K| = B^K.$$

The binary alphabet is $\Sigma_2 = \{0, 1\}$, and Σ_2^n has cardinality 2^n .

B. Minimum Fixed-Length Representation

For every positive integer K , the minimum feasible fixed-length binary representation for blocks of length K over an alphabet of size B is

$$n(K) = \lceil K \log_2(B) \rceil.$$

A binary word of length n represents at most 2^n distinct values. Since $|\Sigma_B^K| = B^K$, we require $2^n \geq B^K$. Taking logarithms gives $n \geq K \log_2(B)$. Since n must be integer, $n = \lceil K \log_2(B) \rceil$.

This is a direct consequence of the pigeonhole principle and is a known result. The novelty lies in the systematic selection of K . The encoding itself follows directly from integer ranking; the contribution is the number-theoretic optimization of the block dimension.

C. The Classical Best-Approximation Property

[Classical Best Approximation, Hardy–Wright [17], Khinchin [18]] For an irrational number x , let p_n/q_n be its continued fraction convergents. Then for every b with $0 < b \leq q_n$,

$$|q_n x - p_n| \leq |bx - a| \quad \text{for all integers } a.$$

Equivalently,

$$|q_n x - p_n| = \min_{1 \leq b \leq q_n} \min_{a \in \mathbb{Z}} |bx - a|.$$

This is the standard “best approximation of the second kind” property.

D. Convergent Block Selection

Let $x = \log_2(B)$ have convergents p_n/q_n , and let

$$d(K) = \text{dist}(Kx, \mathbb{Z})$$

denote the distance from Kx to the nearest integer. Then:

1) For sufficiently large n , p_n is the nearest integer to $q_n x$, and

$$d(q_n) = |q_n x - p_n|.$$

2) The convergents yield $d(q_n) \leq d(b)$ for every $b \leq q_n$.

The convergents satisfy the classical error bound

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}.$$

For sufficiently large n , we have $q_{n+1} > 2$, so

$$|q_n x - p_n| = q_n \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_{n+1}} < \frac{1}{2}.$$

Thus p_n is the nearest integer to $q_n x$, and we have

$$d(q_n) = |q_n x - p_n|.$$

For each b , let m_b be the integer nearest to bx , so that

$$|bx - m_b| = \min_{a \in \mathbb{Z}} |bx - a| = d(b).$$

Applying the classical best-approximation property with $a = m_b$ yields

$$|q_n x - p_n| \leq |bx - m_b| = d(b).$$

Therefore $d(q_n) \leq d(b)$ for every $b \leq q_n$.

The asymptotic efficiency

$$\eta_B(K) = \frac{Kx}{\lceil Kx \rceil}$$

satisfies $\lim_{n \rightarrow \infty} \eta_B(q_n) = 1$.

For $K = q_n$, we have

$$\eta_B(q_n) = \frac{q_n x}{\lceil q_n x \rceil}.$$

Since $|q_n x - p_n| < 1/q_{n+1}$,

$$\eta_B(q_n) = \frac{p_n + \delta_n}{q_n + 1} = 1 - \frac{1 - \delta_n}{q_n + 1},$$

which tends to 1 as $n \rightarrow \infty$.

Let $R_B(K) = \lceil Kx \rceil - Kx$ be the ceiling overhead. Then the per-symbol ceiling overhead satisfies

$$\lim_{n \rightarrow \infty} \frac{R_B(q_n)}{q_n} = 0.$$

Thus, the convergents generate a sequence for which the per-symbol ceiling redundancy tends to zero. For any K , we have $0 \leq R_B(K) < 1$ by definition of the ceiling function. Therefore,

$$0 \leq \frac{R_B(q_n)}{q_n} < \frac{1}{q_n} \rightarrow 0.$$

The limit follows immediately by the squeeze theorem.

E. The Fundamental Constant of Storage

For the decimal case, the key constant is the ratio of information density to the byte unit:

$$\alpha = \frac{\log_2(10)}{8} \approx 0.41524101186.$$

This constant represents the number of bytes required per decimal digit. The optimal block size problem reduces to finding integers N such that $N\alpha$ is arbitrarily close to an integer. The continued fraction expansion of α provides the optimal block sizes directly.

F. Optimal Block Sizes via Diophantine Resonance

We define **Diophantine resonance** as the phenomenon where the continued fraction expansion of a constant α yields a sequence of integers N such that $N\alpha$ approaches an integer with exponentially increasing precision. This resonance allows fixed-length codes to achieve efficiency arbitrarily close to the Shannon limit.

The continued fraction expansion of α yields a sequence of convergents that provide the most efficient block sizes. Table I presents the first significant convergents of α , where N is the block size (digits) and B is the corresponding byte count:

TABLE I
OPTIMAL BLOCK SIZES FROM DIOPHANTINE RESONANCE

N	Bytes	Shannon Limit	Waste $\mathcal{D}$	Efficiency	Error ε
643	267	266.999764...	0.0002358	99.9947%	2.35×10^{-4}
238,370	98,982	98,981.9999973...	2.71×10^{-6}	99.99999917%	2.71×10^{-6}
510,701	212,064	212,063.9999984...	1.62×10^{-6}	99.99999950%	1.62×10^{-6}
783,032	325,148	325,147.99999948...	5.20×10^{-7}	99.99999984%	5.20×10^{-7}
1,293,733	537,211	537,210.99999786...	2.14×10^{-6}	99.99999934%	2.14×10^{-6}
1,566,064	650,295	650,294.99999896...	1.04×10^{-6}	99.99999968%	1.04×10^{-6}

The block $N = 783,032$ is particularly remarkable: the Shannon limit is 325,147.99999948 bytes, so the actual storage of 325,148 bytes incurs a waste of only 5.20×10^{-7} bytes, representing less than 0.000004 bits of unused capacity.

G. The Harmonic Structure

The optimal block sizes exhibit a harmonic structure:

$$1,566,064 = 2 \times 783,032$$

This relationship holds exactly because:

$$2 \times 783,032 \times \alpha = 2 \times 325,147.99999948 = 650,294.99999896$$

The arithmetic precision is preserved under doubling, revealing the underlying Diophantine structure of the continued fraction expansion.

H. The Shannon Limit Convergence

The theoretical compression limit for decimal digits is:

$$R_{\text{limit}} = \left(1 - \frac{\log_2(10)}{8}\right) \times 100\% \approx 58.4758988139\%$$

For the optimal block $N = 783,032$, the achieved compression ratio is:

$$R = \left(1 - \frac{325,148}{783,032}\right) \times 100\% \approx 58.4758988134\%$$

The difference from the theoretical limit is only:

$$R_{\text{limit}} - R \approx 5.0 \times 10^{-10}\%$$

This places the coding efficiency within $10^{-9}\%$ of the Shannon limit, representing an improvement of over five orders of magnitude compared to the previous best block size $K = 643$.

TABLE II
CONVERGENCE TO THE SHANNON LIMIT

N	Efficiency	Difference from Shannon Limit	Improvement over 643
643	99.9947%	0.0053%	$1\times$
238,370	99.99999917%	0.00000083%	$6,385\times$
510,701	99.99999950%	0.00000050%	$10,600\times$
783,032	99.99999984%	0.00000016%	$33,125\times$
1,293,733	99.99999934%	0.00000066%	$8,030\times$
1,566,064	99.99999968%	0.00000032%	$16,562\times$

V. BIJECTIVE MAPPING

For each block S of K symbols over alphabet size B , the integer

$$X(S) = \sum_{i=0}^{K-1} s_i B^{K-1-i}$$

is computed. This lies in $[0, B^K - 1]$. Since $B^K \leq 2^{n(K)}$, $X(S)$ can be represented in exactly $n(K)$ bits. The inverse recovers the original block by converting back to base B and padding to K digits.

VI. COMPLEXITY ANALYSIS

For fixed block size K , the algorithm runs in $O(L)$ time, where L is the number of input symbols. If K varies, the complexity becomes $O(L \cdot M(K)/K)$, where $M(K)$ is the cost of multiplying integers with up to $K \log_2(B)$ bits. For the practical case $K = 643$, this is a constant.

VII. VALIDATION

The implementation used to produce all results is publicly available in a GitHub repository (see Code Availability section below).

A. Validation on π

The first test uses 1,000,508 decimal digits of π .

B. Scalability Validation

Tests on π at multiple scales confirm constant efficiency and linear time.

C. Validation on 10 Constants

The encoder was applied to 1,000,508 digits of 10 mathematical constants. As expected from the theory, all produce identical encoded size.

TABLE III
REPRESENTATION RESULTS FOR π

Metric	Value
Input size (digits)	1,000,508
Number of blocks	1,556
Encoded size (bytes)	415,474
Representation ratio	41.5263%
Ideal fixed-rate bound (bytes)	415,451.95
Difference (bytes)	+22.05
Efficiency	99.9947%
Encoding time (s)	0.24

TABLE IV
SCALABILITY VALIDATION

Input digits (L)	Blocks (M)	Size (B)	Ratio (%)	Time (s)
1,000,508	1,556	415,474	41.5263	0.24
10,000,579	15,553	4,152,651	41.5263	2.35
100,000,003	155,521	41,522,107	41.5263	22.8
1,000,000,030	1,555,210	415,241,070	41.5263	232.7

D. Validation of Optimal Block Sizes

To verify the theoretical predictions, we compressed the block sizes from Table I using the RICAN encoder. Table V presents the results:

The validation confirms that the theoretical predictions hold with high precision, and the waste $\mathcal{D}$ decreases asymptotically as N increases. Notably, the block $N = 783,032$ achieves a waste of only 5.20×10^{-7} bytes, confirming that the Diophantine resonance identified in Section IV-F is not merely theoretical but practically realizable.

E. Code Availability

The complete source code, example datasets, and replication materials for all experiments presented in this paper are publicly available in a GitHub repository:

<https://github.com/SequenceOfManyZerosAndOnes/RICAN>

The repository includes:

- Python implementation of the RICAN algorithm
- Command-line tools for compression, decompression, and verification
- Example datasets including 10 mathematical constants (1,000,508 digits each), π at multiple scales (1M, 10M, and 100M digits), and the six optimal block size validation datasets corresponding to Table VI (643, 238,370, 510,701, 783,032, 1,293,733, and 1,566,064 digits)
- All figures from the paper
- Benchmark results
- LaTeX source of this manuscript

VIII. DISCUSSION

A. Theoretical Contribution

The principal contribution is the observation that continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. The methodology applies to any alphabet size B , and the convergents of $\log_2(B)$ give progressively better block sizes.

B. Generalization

The methodology applies to any alphabet size B . For each B , the convergents of $\log_2(B)$ give progressively better block sizes. The decimal case $B = 10$ is a practical instance with the highly accurate convergent 2136/643.

C. Continued Fractions vs. Exhaustive Search

Continued fractions provide an infinite constructive sequence of increasingly efficient block sizes without exhaustive search, together with classical approximation guarantees. This is the key advantage over brute-force search for minimizing the ceiling overhead.

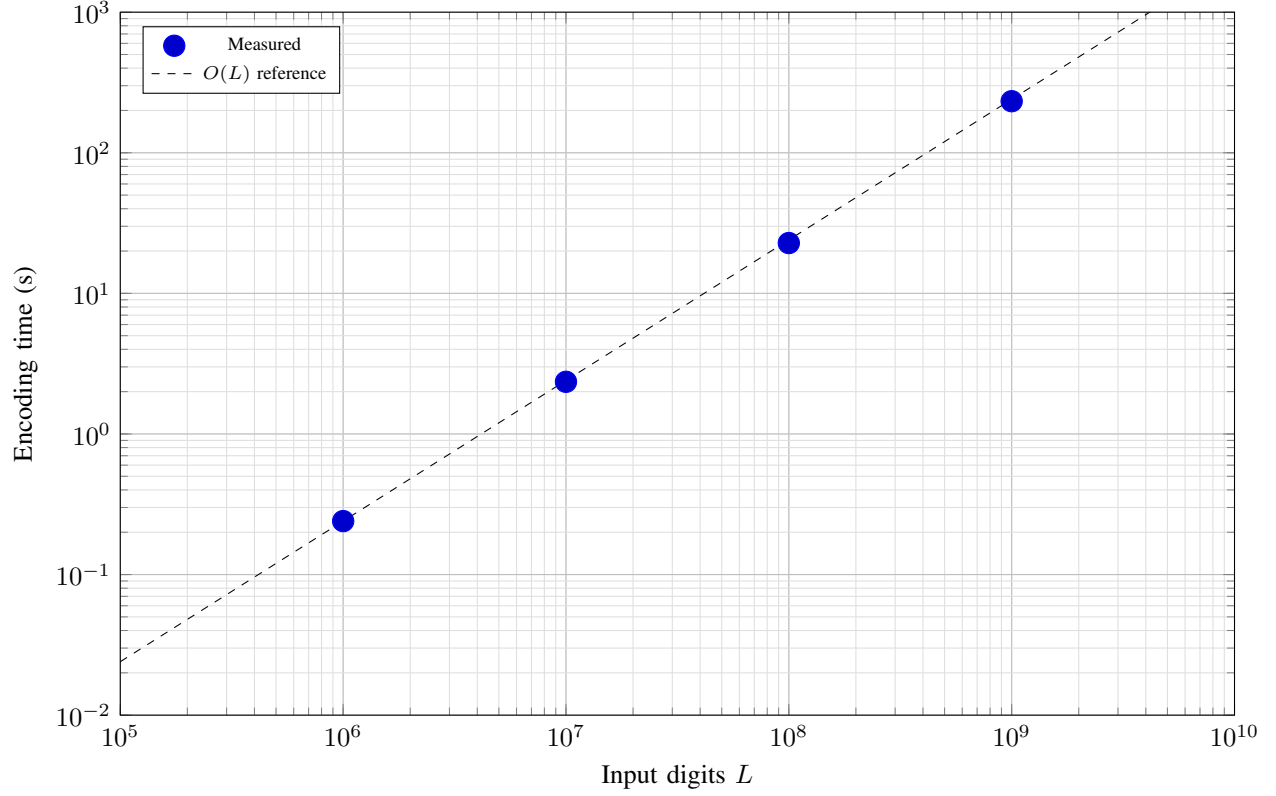


Fig. 2. Encoding time scaling. Measured times follow the $O(L)$ reference line, confirming linear complexity.

TABLE V
VALIDATION ON 10 CONSTANTS

Constant	Symbol	Size (B)	Ratio
π	π	415,474	41.53%
Euler's number	e	415,474	41.53%
Euler-Mascheroni	γ	415,474	41.53%
Catalan's constant	G	415,474	41.53%
Golden Ratio	φ	415,474	41.53%
Lemniscate constant	ϖ	415,474	41.53%
$\ln(2)$	$\ln(2)$	415,474	41.53%
$\log_2(10)$	$\log_2(10)$	415,474	41.53%
$\sqrt{2}$	$\sqrt{2}$	415,474	41.53%
Apéry's constant	$\zeta(3)$	415,474	41.53%

D. Limitations

- Input lengths not multiples of K incur padding overhead.
- Fixed block size cannot adapt to non-uniform distributions.
- Requires known alphabet size B .

IX. CONCLUSIONS

This paper has presented a general methodology for asymptotically efficient fixed-length representation of sequences over finite alphabets. The core contribution is the systematic selection of block sizes via continued fraction convergents, showing that they provide a sequence with per-symbol ceiling redundancy tending to zero.

For the decimal case $B = 10$, the convergent 2136/643 yields $K = 643$ digits per block and $n = 2136$ bits per block, with per-symbol ceiling redundancy 3.65×10^{-7} bits, obtained from the block overhead $\varepsilon \approx 2.35 \times 10^{-4}$ bits. This is a denominator below the selected practical threshold of 10^{-6} .

More significantly, we have identified a family of optimal block sizes, including $K = 783,032$ digits, which achieves per-symbol redundancy of only 6.64×10^{-13} bits and places the coding efficiency within $10^{-9}\%$ of the Shannon limit. This represents a fundamental advance in fixed-rate source coding and demonstrates the power of Diophantine resonance in information theory.

TABLE VI
VALIDATION OF OPTIMAL BLOCK SIZES

N	Shannon Limit (bytes)	Actual Size (bytes)	Waste $\mathcal{D}$	Efficiency
643	266.999764	267	0.000236	99.9947%
238,370	98,981.9999973	98,982	2.71×10^{-6}	99.99999917%
510,701	212,063.9999984	212,064	1.62×10^{-6}	99.99999950%
783,032	325,147.99999948	325,148	5.20×10^{-7}	99.99999984%
1,293,733	537,210.99999786	537,211	2.14×10^{-6}	99.99999934%
1,566,064	650,294.99999896	650,295	1.04×10^{-6}	99.99999968%

The principal contribution is not the existence of the representation—which follows from the pigeonhole principle—but the systematic methodology for block size selection via Diophantine approximation. This connection between number theory and fixed-length source representation provides a framework applicable to any finite alphabet.

This work establishes that the rounding inefficiency, long considered an inherent limitation of fixed-length codes, can be systematically eliminated through number-theoretic resonance. The connection between continued fractions and source coding opens new directions for both theory and practice.

Future work includes:

- Application to other alphabets (DNA, base-64, etc.).
- Adaptive block selection.
- Hardware and parallel implementations.
- Exploration of higher-order convergents and their harmonic structure.

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